

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

A **SYN flood attack**, which is a type of **Denial-of-Service (DoS)** attack.

The logs show that:

A large number of **TCP SYN requests** are being sent from the same unfamiliar IP address to the company's web server. The server becomes overwhelmed and starts having difficulty responding to legitimate employee requests. The logs also show **RST, ACK packets** and **HTTP 504 Gateway Time-out** errors.

This event could be:

A **network-level DoS attack (SYN flood)** because one IP address is repeatedly flooding the web server with SYN requests. The log specifically identifies this pattern as a SYN flood attack.

Section 2: Explain how the attack is causing the website to malfunction

Three-way TCP handshake

1. SYN:

The visitor sends a **SYN** packet to the web server to request a connection. SYN means "synchronize."

2. SYN-ACK:

The web server sends a **SYN-ACK** packet back to the visitor, agreeing to the connection. The server reserves resources for the connection.

3. ACK:

The visitor sends an **ACK** packet back to the server. This completes the three-way handshake and establishes the TCP connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

The malicious actor **floods the web server with a large number of SYN requests**. The server tries to process these requests and uses its resources to handle them. When the number of SYN requests becomes greater than the server's available resources, the server becomes overwhelmed and cannot properly respond to legitimate visitors.

Explain what the logs indicate and how that affects the server:

The logs show that the attacker, **203.0.113.0**, repeatedly sends SYN packets to the web server **192.0.2.1**. The requests continue at a rapid rate. As the attack continues, legitimate visitors begin receiving **RST, ACK** packets and **HTTP 504 Gateway Time-out** errors. Eventually, the web server stops responding to legitimate employee traffic.